

Wrong Shape Does Not Mean Biased

Topic 5.5 · TODAY'S PROBLEM

Suppose a video platform has 5,000 active accounts and says that $p = 0.08$ of them regularly use captions. An SRS of 100 accounts, selected without replacement, contains 14 regular caption users, so $\hat{p} = 0.14$.

Jules: “Because $n = 100 > 30$, $\hat{p}$ is approximately normal. I get $P(\hat{p} \geq 0.14) = 0.0135$, which proves caption use increased.”

Nora: “Because $np = 8 < 10$, a normal model is not justified. Therefore $\hat{p}$ is biased and this sample tells us nothing.”

Caption-use sample

Source	Accounts	Users	Rate
Population	5,000	400	0.08
SRS	100	14	0.14

FIRST TAKE (5 min, on your sheet)

Whose reasoning is closer to correct, Jules's or Nora's? Name one condition or claim you would check.

- DOK 1** (a) Identify the population parameter, sample statistic, population size, and sample size. Verify the reported value of $\hat{p}$.
- DOK 2** (b) Find the mean and standard deviation of the sampling distribution of $\hat{p}$. Check the 10 percent condition and both Large Counts values.
- DOK 3** (c) Adjudicate both claims. Use a binomial calculation or valid simulation for $P(\hat{p} \geq 0.14)$ under $p = 0.08$. State the strongest conclusion warranted, explain why failed normality does not create bias, and give the consequence of each overclaim.

Video + follow-along: u5_lesson5_live.html (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

